HDOC Day-48

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Write a C program that reads the size and elements of a variable-length integer array, arr[]. Pass the array to a function named pivot_index() using its address. Use pointers to pass the array between the calling function and pivot_index(). The function pivot_index() must:

- Calculate the left sum (sum of all elements strictly to the left of each index).
- Calculate the right sum (sum of all elements strictly to the right of each index).
- Treat the left sum = 0 for the first element.
- Treat the right sum = 0 for the last element.
- Find and display the leftmost index at which left sum = right sum.
- Display -1 if no pivot index exists.

1. Problem Understanding

The program reads the size and elements of a variable-length integer array. The array is passed to pivot_index() using its address. For every index, the function compares the sum of elements on its left with the sum of elements on its right. It returns the leftmost index where both sums are equal; otherwise, it returns -1.

2. Standard Test Cases

Test Case	Size (n)	Array Input	Expected Output
1	6	1 7 3 6 5 6	Pivot index: 3
2	3	1 2 3	-1
3	3	2 1 -1	Pivot index: 0
4	1	5	Pivot index: 0
5	5	1 -1 0 0 0	Pivot index: 0

3. Algorithm

Main Algorithm

1. Start.
2. Read the size n.
3. Declare a variable-length integer array arr[n].
4. Read n array elements.
5. Pass the array address and size to pivot_index().
6. Display the pivot index returned by the function.
7. Stop.

Function: pivot_index()

1. Calculate the total sum of all array elements using the pointer.
2. Initialize leftSum to 0.
3. Traverse the array from index 0 to n - 1.
4. For each index, calculate rightSum = totalSum - leftSum - current element.
5. If leftSum equals rightSum, return the current index immediately.
6. Add the current element to leftSum and continue.
7. If no such index is found, return -1.

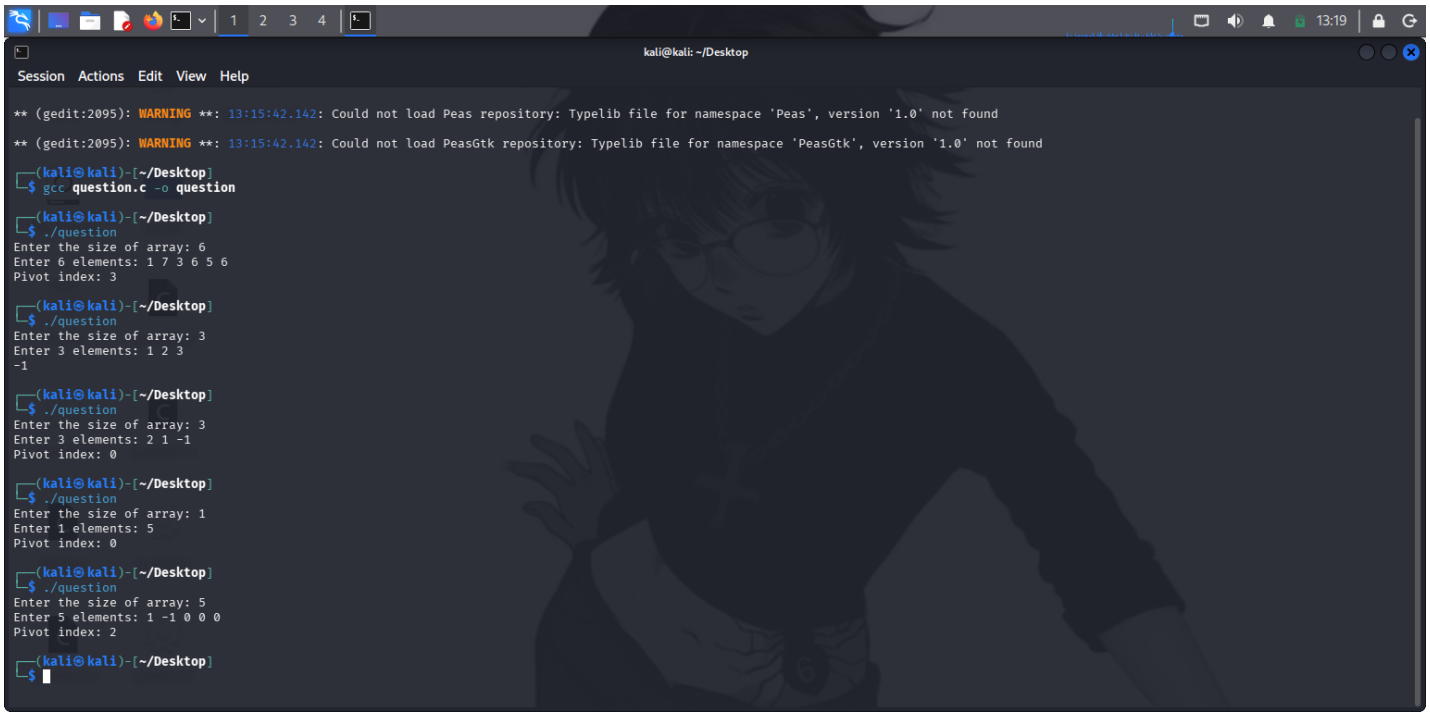
4. C Program

```
#include <stdio.h>
int pivot_index(int *arr, int n)
{
   int totalSum = 0, leftSum = 0;
   int i, rightSum;
   for(i = 0; i < n; i++)
   {
      totalSum += *(arr + i);
   }
   for(i = 0; i < n; i++)
   {
      rightSum = totalSum - leftSum - *(arr + i);

      if(leftSum == rightSum)
      {
         return i;
      }
      leftSum += *(arr + i);
   }
   return -1;
}
int main()
{
   int n, i, result;
   printf("Enter the size of array: ");
   scanf("%d", &n);
   int arr[n];
   printf("Enter %d elements: ", n);
   for(i = 0; i < n; i++)
   {
      scanf("%d", &arr[i]);
   }
   result = pivot_index(arr, n);
   if(result == -1)
      printf("-1\n");
```

```
    else
        printf("Pivot index: %d\n", result);
    return 0;
}
```

5. Compilation and Execution Evidence



6. Verification of All Test Cases

Test Case	Size (n)	Array Input	Actual Result	Status
1	6	1 7 3 6 5 6	Pivot index: 3	Pass
2	3	1 2 3	-1	Pass
3	3	2 1 -1	Pivot index: 0	Pass
4	1	5	Pivot index: 0	Pass
5	5	1 -1 0 0 0	Pivot index: 0	Pass